VISHNU DHARSAN A S

+91 7502422244 | vishnudharsan1@gmail.com | LinkedIn: linkedin.com/in/vishnu-dharsan-773632305

SUMMARY

Data-focused Software Engineer with 1+ year of experience in Python, SQL, and machine learning, building data pipelines, analytics, and AI solutions (NLP, RAG). Skilled in data cleaning and preparation, exploratory analysis, Power BI dashboards, and deep learning models, including a published bearing fault classification paper. Seeking Data Analyst or Data Scientist roles.

TECHNICAL SKILLS

Programming & Querying: Python, SQL, C, Java, JavaScript

Data Analysis & Visualization: Pandas, NumPy, Power BI, MS Excel, Matplotlib, Seaborn, Plotly

Machine Learning & Deep Learning: Scikit-learn, TensorFlow, Keras, LightGBM, SVM, CNN, ANN, Hugging Face Transformers

NLP & Generative AI: LangChain, Retrieval-Augmented Generation (RAG), Ollama, OpenAI API, Embeddings, Semantic Search, FAISS, ChromaDB, Prompt Engineering

Databases: SQL Server, PostgreSQL, MySQL, MongoDB

Tools & Platforms: Jupyter Notebook, Git, GitHub, Linux, Docker, Podman, Google Cloud Platform (GCP), FastAPI, Flask, REST APIs

Concepts: Data Cleaning & Preparation, Exploratory Data Analysis, Feature Engineering, Data Warehousing, Model Evaluation, Data Structures & Algorithms, Object-Oriented Programming (OOP)

EXPERIENCE

ASM Technologies (Software Engineer) | Aug 2025 – Present

- Built an end-to-end Retrieval-Augmented Generation (RAG) assistant for industrial failure analysis, supporting troubleshooting, log analysis, and failure analysis (FA) report generation.
- Built data ingestion and preparation pipelines to clean, chunk, and index 1,000+ technical documents and logs for semantic retrieval using Ollama LLMs, ChromaDB, and LangChain.
- Analyzed equipment logs to support root cause identification, and evaluated retrieval and prompt quality to reduce hallucinations by ~40%, cutting manual FA report preparation time by ~60%.
- Developed REST APIs using FastAPI and Flask with PostgreSQL and MongoDB for data storage, chat history, and workflows; reduced average API response latency by ~35% through async endpoints, query optimization, and caching.
- Containerized and deployed services using Podman/Docker for reproducible deployments.

AES Technologies (Project Intern) | Oct 2023 – May 2024

HRIQ – HRMS System

- Analyzed employee data using SQL and .NET to identify trends and improve reimbursement processing efficiency.
- Developed the Reimbursement feature of a full-stack HRMS application for employee management.
- Collaborated with a cross-functional team of 6, using jQuery, C#, and SQL Server to deliver a scalable HR platform, decreasing feature deployment time by 20%.

PROJECTS

HR Data Analytics & Marketing Analysis (Python, SQL Server, Power BI, Pandas, NumPy)

- Analyzed HR data to derive actionable insights on employee performance, retention, and workforce optimization, using data analysis, warehousing, and visualization.
- Analyzed shop data to understand customer behavior and helped create a better business strategy.

Bearing Fault Classification (Paper Published)

- Developed a deep learning model to predict faults and remaining lifetime of bearings used in machines, including data preprocessing and feature preparation.
- *Tech:* Python, CNN, ANN, TensorFlow, SVM, LightGBM.

Automotive Failure Detection and Recovery Recommendation

- Created a hybrid LLM + rules-based system that classified automotive failures with ~90% accuracy across 4 OEM datasets, generated recovery recommendations, and built a web UI.
- Compared outputs of different LLMs to select the best-suited model for performance.
- *Tech:* Python, LLMs (GPT-3, Llama 3, Bard/Gemini, Claude, Falcon), rule-based classification, React, Node.js.

EDUCATION

PSG College of Engineering, Coimbatore, TN

Bachelor of Engineering – Computer Science and Engineering | CGPA: 8.0 | Nov 2021 – Apr 2025

CS Academy, Erode, TN

Class XII – 94% (May 2021) | Class X – 92% (May 2019)

ACHIEVEMENTS & ACTIVITIES

- Represented ASM Technologies at CII INNOVERGE 2025.
- Delivered a technical talk: "Building RAG Chatbot using NVIDIA NIM".
- Secretary, Entrepreneur's Club (event organisation, speaker sessions, overall management); Photography and Publicity Head, CSE Association; Class Representative (5th semester).